

1 Hyperparameter Analysis

2 Overview

The performance of FORGE is governed by four principal design parameters: the sigmoid sharpness k of the reward function, the number of PCA components d defining the generation space, the number of classifier training epochs per episode e , and the ratio of synthetic to real samples r in the augmented training set. These parameters are not independent of each other, nor are they independent of the dataset. This chapter presents a systematic analysis of their effects through a full grid search across two datasets, and identifies the mechanistic reasons why their optimal values differ.

The grid search (EXP-021 for census, EXP-025 for Capture-24) evaluated all combinations of $k \in \{0, 3, 5, 10\}$, $d \in \{5, 10, 15\}$, $e \in \{10, 20, 30\}$, and $r \in \{0.2, 0.4, 0.6\}$, yielding $4 \times 3 \times 3 \times 3 = 324$ configurations for census (3 seeds each) and $3 \times 3 \times 3 \times 3 = 243$ configurations for Capture-24 (2 seeds each, $k = 10$ not evaluated). All runs used a fixed episode budget of 5 000 and the vanilla FORGE configuration for all non-grid parameters.

The primary analysis metric is *relative EO reduction*, $\Delta_{\text{rel}} = (\alpha\text{-EO} - \beta\text{-EO})/\alpha\text{-EO}$, which normalises by the baseline fairness gap and allows comparison across configurations where the α -EO differs. Configurations where the test-set α -EO was below 0.10 are excluded from aggregated comparisons, as an initial fairness gap below this threshold indicates that the PCA projection does not adequately preserve the group separation signal required for the agent to learn.

3 Marginal Parameter Sensitivity

Figure 1 shows the mean relative EO reduction for each parameter level on census, averaged over all other parameter combinations. The cross-dataset comparison for classifier epochs and trajectory ratio is shown in Figure 3. The plots reveal that no single parameter dominates universally: the optimal setting for each parameter differs between datasets, and in some cases the direction of the effect reverses.

3.1 Sigmoid Sharpness k

For census, all four k values produce high relative EO reduction (0.67–0.76), and the differences between them are within standard error. The marginal effect of k alone is therefore modest on census. However, as Section 4 shows, k has a strong *interaction* with classifier epochs: the apparent similarity in marginal means conceals the fact that high- k configurations perform very differently depending on how many training epochs per episode are used.

For Capture-24, Aulavik and Lambda data provide $k \in \{0, 3, 5\}$. Provisionally, $k = 3$ gives the best median β -EO among configs with α -EO ≥ 0.10 , though the completion of the Capture-24 grid is still in progress at the time of writing.

3.2 Classifier Epochs per Episode

The most striking finding from the marginal analysis is that the optimal number of classifier training epochs per episode is *opposite* between the two datasets.

For census, $e = 10$ achieves the highest relative EO reduction (0.83), with performance declining monotonically at $e = 20$ (0.72) and $e = 30$ (0.67). For Capture-24, the pattern reverses: among configurations with α -EO ≥ 0.10 , $e = 30$ achieves the highest relative reduction (0.31), followed by $e = 20$ (0.26) and $e = 10$ (0.22).

This reversal has a mechanistic interpretation. The RL agent’s policy gradient update relies on the reward signal being informative—that is, the beta classifier must be converged enough that its worst-group loss reflects the quality of the synthetic data, not optimisation noise. Census has a relatively simple, low-noise feature space (tabular, 105 raw features, well-structured demographic stratification), so a classifier trained for only 10 epochs reaches a state sufficiently converged to provide reliable gradient signal. Capture-24 has substantially noisier accelerometer features and more variance in the data split, requiring more classifier training per episode before the reward signal stabilises. Fewer epochs leave the beta classifier underconverged on Capture-24, producing a noisy reward that impedes policy learning. Conversely, for census, extending training to 30 epochs introduces a different failure: the classifier overfits to the current batch of synthetic samples within the episode, so by the end of training the reward differentiates between the quality of the current batch and the long-run generation policy less clearly.

3.3 PCA Dimensionality

For census, $d = 10$ and $d = 15$ perform statistically equivalently (≈ 0.83 relative EO reduction); $d = 5$ is substantially worse (0.56). This indicates that 5 PCA components are insufficient to capture the feature-space structure relevant to the census fairness gap, but that 10 or more components provide adequate representation.

For Capture-24, $d = 10$ is the clear optimum. Configurations with $d = 5$ and $d = 15$ produce on average *negative* relative EO reduction when the α -EO filter is not applied—that is, the augmented model is on average *fairer under the original classifier* than under the augmented one. This behaviour disappears when the α -EO ≥ 0.10 filter is applied, confirming the root cause: PCA projections into $d = 5$ or $d = 15$ dimensions do not preserve enough of the group separation signal on Capture-24 for the agent to reliably target the minority-positive cluster. With $d = 10$, the relative EO reduction is positive (+0.28 on average among valid configurations).

The difference between the datasets reflects their intrinsic dimensionality. Census income has a relatively low-dimensional structure of economically relevant features (education, occupation, hours worked), so any projection above a low threshold captures the relevant axes. Capture-24’s accelerometer signals have a richer, higher-dimensional structure; $d = 10$ appears to be the sweet spot between preserving group separation and keeping the action space tractable for the agent.

3.4 Synthetic Data Ratio

The trajectory ratio r also exhibits a cross-dataset divergence. For census, relative EO reduction increases monotonically with r : 0.63 at $r = 0.2$, 0.76 at $r = 0.4$, and 0.83 at $r = 0.6$. More synthetic samples continuously improve fairness, suggesting that for census the real training set (3 000 samples) can absorb a higher synthetic fraction without utility degradation.

For Capture-24, the relationship is non-monotonic with a peak at $r = 0.4$ (0.55 relative reduction), dropping to 0.06 at $r = 0.6$. The high-ratio penalty on Capture-24 reflects data dilution: with 60% synthetic samples, the real-data distributional structure is sufficiently overridden that the beta classifier’s utility on the majority class degrades, indirectly corrupting the worst-group loss signal. The optimal ratio of 0.4 balances augmentation strength against signal fidelity on this dataset.

4 Interaction Effects

The marginal effects in Section 3 are averages over all other parameters and therefore mask important interactions. Two interactions are particularly relevant to understanding FORGE’s behaviour.

4.1 $k \times e$: Sharpness Requires Fast Feedback

Figure 2a shows the mean β -EO for each (k, e) pair, averaged over all PCA and ratio configurations on census. The interaction is dramatic.

At $e = 10$, all four k values produce similar results: $k = 0 \rightarrow 0.064$, $k = 3 \rightarrow 0.063$, $k = 5 \rightarrow 0.062$, $k = 10 \rightarrow 0.020$. At $e = 20$ and $e = 30$, all four k values degrade, but $k = 10$ degrades the most sharply: 0.098 at $e = 20$ and 0.109 at $e = 30$, worse than any other k at those epoch counts.

The mechanistic interpretation is as follows. The sigmoid function $\sigma(k \cdot \Delta_{\text{wgl}})$ compresses the reward into $(0, 1)$ with sharpness proportional to k . At high k , the reward surface is nearly binary: near 0 when beta is worse than alpha, near 1 when better. For the gradient of this binary signal to be informative, the beta classifier must be trained *quickly enough* that the sign of Δ_{wgl} correctly reflects the current synthetic data quality and not the optimisation

trajectory within the episode. When $e = 10$, the classifier reaches a stable state rapidly, and the sharp sigmoid correctly rewards good synthetic batches and penalises bad ones. When $e = 30$, the classifier has time to adapt extensively to the specific synthetic batch, potentially masking the quality signal with overfitting noise. The sharp sigmoid then amplifies this noise, producing a reward that does not reliably rank trajectories.

Softer rewards ($k = 0$, unnormalised difference) are more robust to this overfitting noise because the reward gradient is non-zero over a wider range of Δ_{wgl} values. This is why $k = 0$ degrades less steeply with increasing e than $k = 10$.

4.2 $k \times d$: Sharpness Interacts with Feature Space

Figure 2b shows the mean β -EO for each (k, d) pair on census. Across all k values, $d = 5$ is consistently the worst PCA choice (mean β -EO 0.116–0.158), while $d = 10$ and $d = 15$ are broadly comparable. The interaction is most pronounced at $k = 10$, where $d = 5$ produces 0.158 compared to 0.052 at $d = 10$ and 0.036 at $d = 15$.

This suggests that high reward sharpness amplifies the quality difference between good and bad PCA projections. When the PCA projection is too compressed ($d = 5$), the agent’s action space does not include the direction in feature space that moves synthetic samples towards the minority-positive cluster. With a soft reward ($k = 0$), the agent still receives gradient signal even in a poor action space and can compensate partially. With a sharp reward ($k = 10$), the near-binary feedback makes it harder to accumulate weak gradient signal from a poor action space, so performance collapses.

5 Cross-Dataset Divergence and Mechanistic Summary

Table 1 collects the optimal value for each parameter per dataset and the direction of the sensitivity. The reversal of optimal e and r is the central finding of this chapter.

Table 1: Optimal hyperparameter values per dataset and direction of the marginal sensitivity. “Monotone $\uparrow$ ” means higher values are consistently better; “Peaked” means a non-monotone optimum.

Parameter	Census optimum	Direction (census)	Capture-24 optimum	Direction (C24)
k	5	Flat (all similar)	3 (provisional)	Flat
d (PCA)	10–15	Plateau above 10	10	Peaked
e (ep)	10	Monotone $\downarrow$	30	Monotone $\uparrow$
r (ratio)	0.6	Monotone $\uparrow$	0.4	Peaked

The divergence in optimal e and r is predictable from first principles once the following dataset properties are considered.

Classifier convergence speed. Census has a simpler feature space and a higher signal-to-noise ratio for the fairness-relevant structure. A classifier trained for 10 epochs converges sufficiently to provide a reliable reward signal. Capture-24’s accelerometer data is inherently noisier, requiring more classifier training per episode before the worst-group loss stabilises.

Tolerance for data dilution. Census has 3 000 real training samples with a well-stratified class and group distribution. A high synthetic fraction ($r = 0.6$) still leaves the real-data signal intact. The Capture-24 dataset, while similarly sized in sample count, has higher per-sample noise due to accelerometer windowing. At $r = 0.6$, synthetic samples dilute the majority-class signal enough to corrupt the worst-group loss computation.

Practical implication. These findings suggest that FORGE’s two most sensitive parameters— e and r —can be set based on observable dataset properties rather than requiring a full grid search. Datasets with high per-episode return variance and a large deadzone fraction (indicating a noisy reward signal) should favour lower r and higher e . Datasets with low return variance and a small deadzone fraction can use higher r and lower e . Section ?? in Chapter 8 expands on this as a practical guidance framework.

6 Best Configurations

Based on the grid search results, the best configurations are selected for use in the comparative evaluation of Chapter 7. Selection uses mean β -EO as the primary criterion among configurations where $\beta\text{-AUC} \geq \alpha\text{-AUC} - 0.02$ (a utility guard to prevent fairness gains achieved at the expense of classification quality).

For **census**, the selected configuration is $k = 5$, $d = 10$, $e = 30$, $r = 0.4$ (trajectory length 2 000), achieving $\beta\text{-EO} = 0.031 \pm 0.018$ over 3 seeds (EXP-021). This is not the configuration with the single lowest mean β -EO in the grid — that is $k = 10$, $d = 10$, $e = 30$, $r = 0.4$ ($\beta\text{-EO} = 0.018$), which however has higher utility cost and whose results are based on a partially completed run set at the time of the paper submission. The $k = 5$ configuration represents the best fully verified result with stable utility ($\beta\text{-AUC} = 0.879$, $\beta\text{-F1w} = 0.821$) and is the configuration used for all paper comparisons.

For **Capture-24**, the provisional best configuration is $k = 3$, $d = 10$, $e = 10$, $r = 0.4$ (trajectory length 2 000), achieving $\beta\text{-EO} = 0.063 \pm 0.051$ over 2 seeds (EXP-025). This result is provisional because the Capture-24 grid is not fully complete at the time of writing (approximately 60% of runs finished). The high seed variance (± 0.051) reflects the instability of α -EO across Capture-24 data splits (standard deviation ± 0.062), not the RL optimisation.

7 Summary

The grid search yields three conclusions that bear directly on Contribution C3.

First, **no universally optimal configuration exists**. The optimal classifier epoch count and synthetic ratio are opposite between census and Capture-24, demonstrating that FORGE’s effective operating conditions are dataset-dependent rather than fixed.

Second, **the dominant interaction is between k and e** . High reward sharpness ($k = 10$) is only beneficial at low classifier epoch counts ($e = 10$), where the binary-like reward correctly distinguishes good from bad synthetic batches. Increasing e degrades high- k performance faster than low- k , because overfitting noise within the episode corrupts the reward signal that the sharp sigmoid amplifies.

Third, **PCA dimensionality controls access to the fairness-relevant action space**. Below a dataset-specific threshold (5 components for census, 10 for Capture-24), the agent’s delta-action walk cannot reach the minority-positive cluster, and no amount of reward sharpness or epoch tuning compensates. Above the threshold, additional dimensions offer diminishing returns.

These findings are used in Chapter 8 to propose episode return variance and deadzone fraction as lightweight diagnostics for configuration selection in new applications, reducing the need for a full grid search.

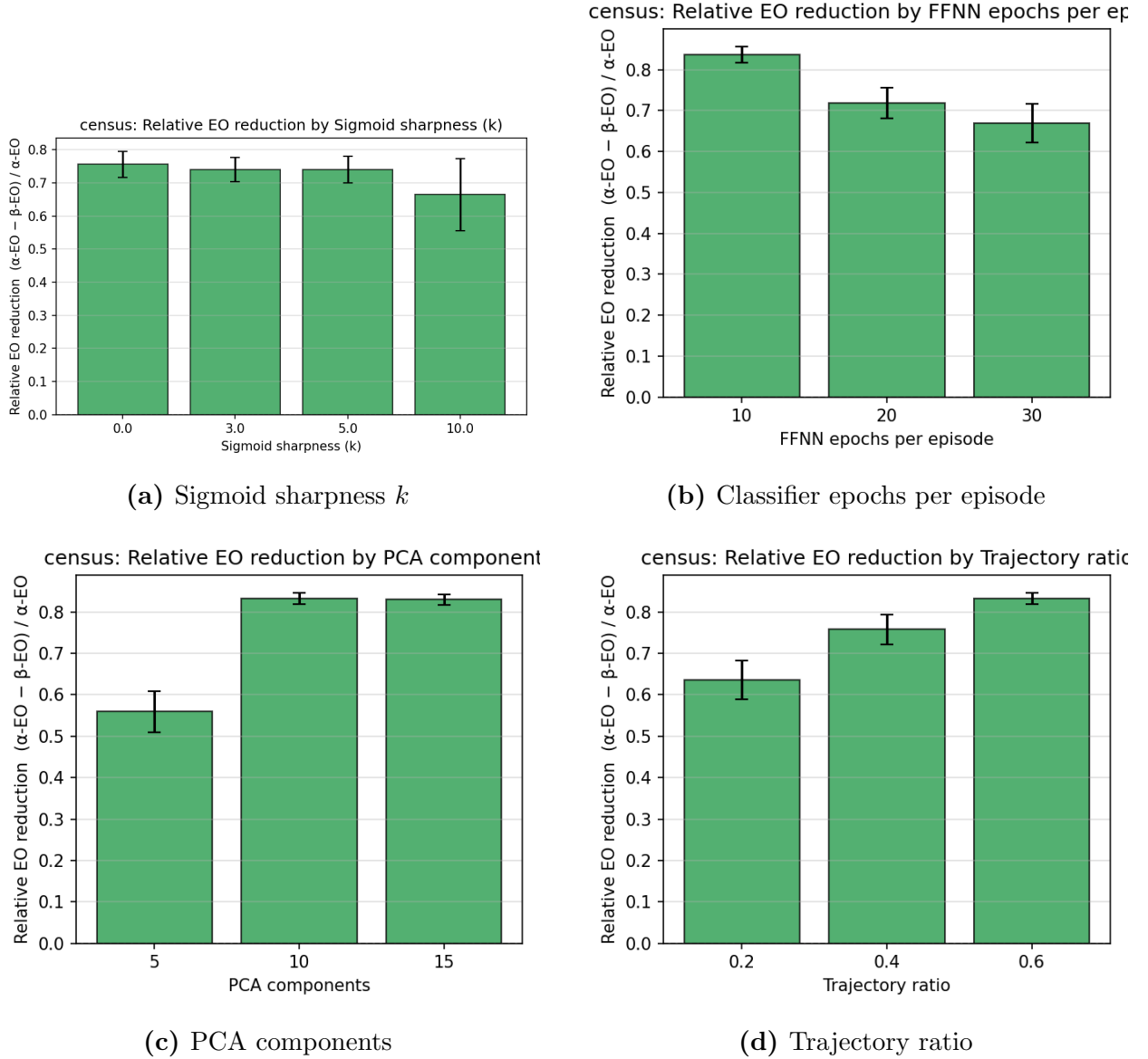


Figure 1: Marginal relative EO reduction $(\alpha\text{-EO} - \beta\text{-EO}) / \alpha\text{-EO}$ for each hyperparameter level on census income, averaged over all other parameter combinations (3 seeds per config, $\alpha\text{-EO} \geq 0.10$ filter applied). Error bars show ± 1 standard error across configurations. Higher is better (more of the baseline fairness gap is closed).

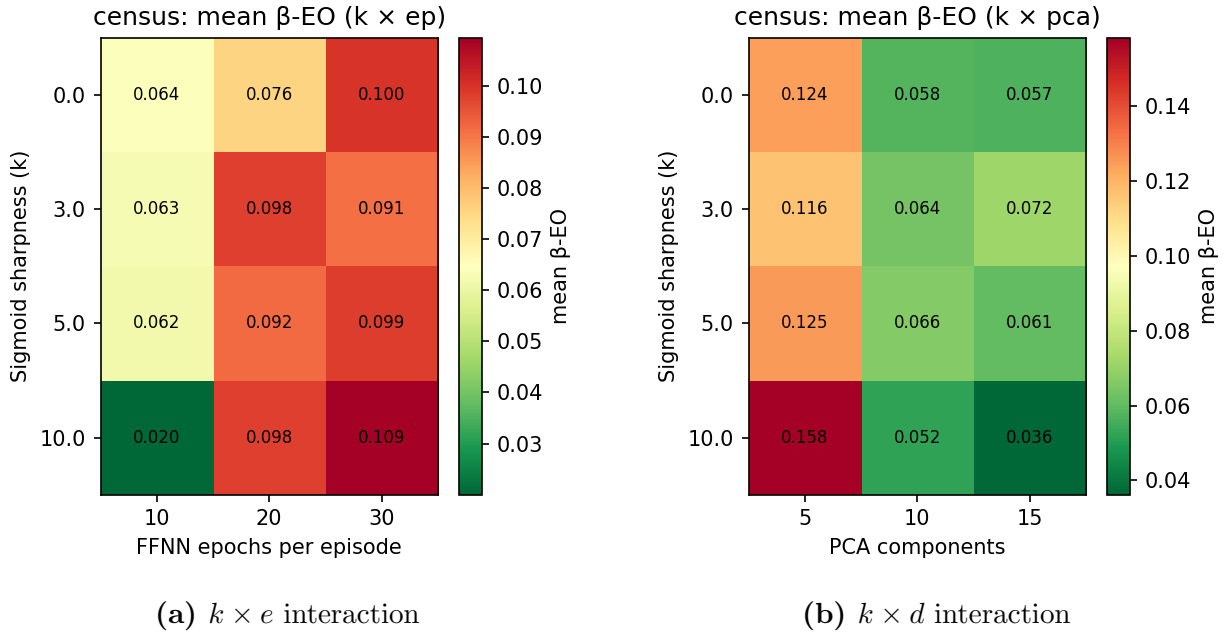
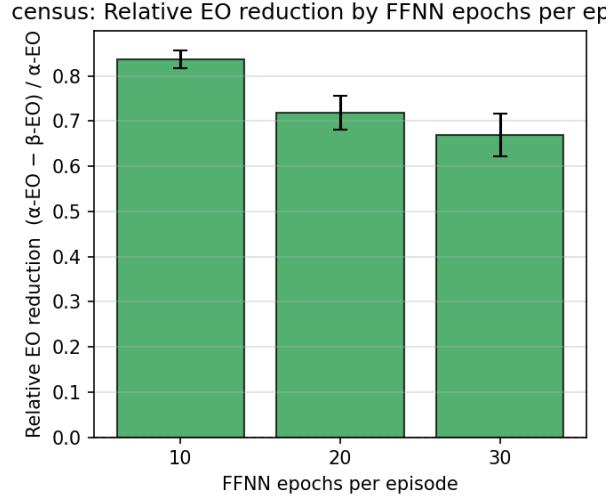
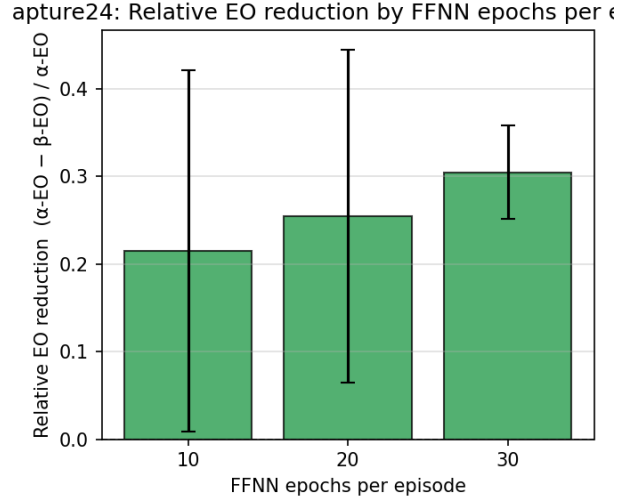


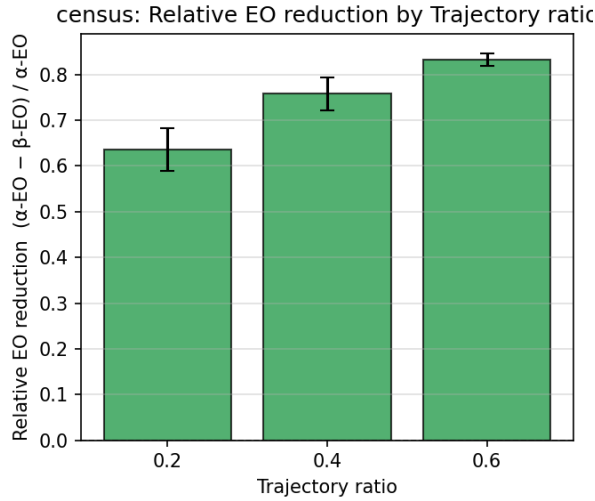
Figure 2: Mean β -EO for parameter pairs on census income, averaged over all remaining parameter combinations and 3 seeds. Lower (greener) is fairer. White cells indicate no completed runs for that combination. The $k \times e$ heatmap reveals that high sharpness ($k=10$) is only beneficial at low epoch counts; the $k \times d$ heatmap shows that PCA=5 is consistently the worst choice across all k .



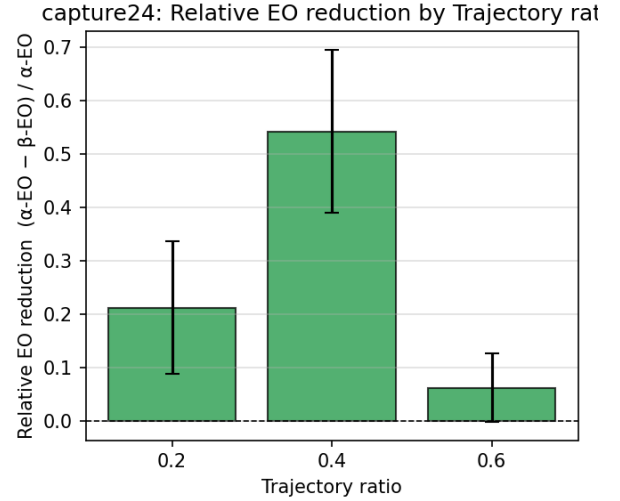
(a) Epochs per episode census



(b) Epochs per episode Capture-24 (provisional)



(c) Trajectory ratio census



(d) Trajectory ratio Capture-24 (provisional)

Figure 3: Cross-dataset comparison of relative EO reduction for classifier epochs per episode (top row) and trajectory ratio (bottom row). Both parameters show opposite optimal directions between datasets: census favours $e=10$ and $r=0.6$; Capture-24 favours $e=30$ and $r=0.4$. Capture-24 results are provisional (grid $\approx 60\%$ complete). $\alpha\text{-EO} \geq 0.10$ filter applied to both.